

MAGNETOM

Replacement of Parts System

Control

Harmony syngo MR
Symphony syngo MR
Sonata syngo MR
Trio
Upgrades to syngo MR:
from Impact/Expert
from Vision

Document Version

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1.1 General

This part contains the exchange procedures for the FRU's and the necessary adjustment procedures. Please observe normal safety precautions in connection with the particular area in which you are working and the overall safety precautions in general.



Protective earth connection test.

If a protective earth connection has to be disconnected for replacing a component, measure the protective earth connection resistance $\leq 100 \text{ m}\Omega$ with a ground test meter (Fluke 1503, or equivalent) between the protective earth connector on the "sub-component" and the protective earth wire connected to the "main component".

Document the successful measurement in the service report (measured value and measured equipment) as required by IEC 62353.

1.2 MPCU1-Box



The replacement requires approximately 30 minutes.

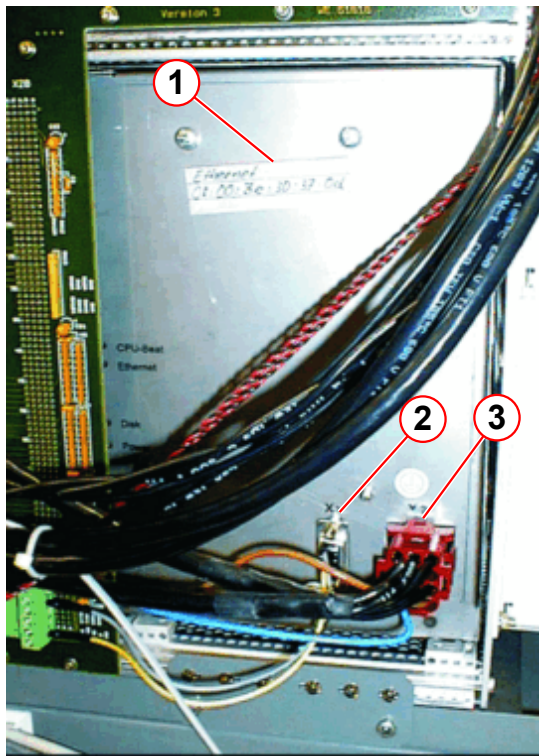
1.2.1 Preliminary Steps

- Shut down NUMARIS
- Switch off the system at the alarm box.

1.2.2 Removing old MPCU1-BOX

- First disconnect at **rear side** (→ Fig. 1 Page 5):
 - **X1**: MPCU Fan Control
 - **X2**: Power supply

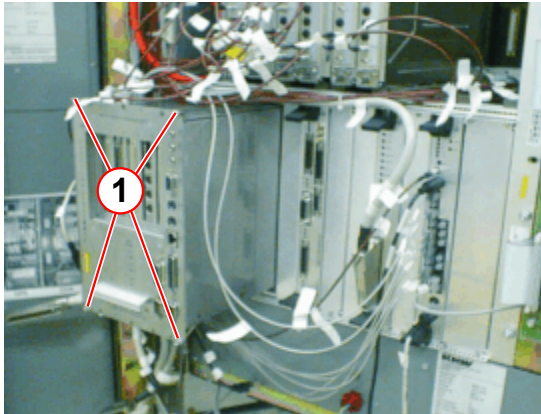
Fig. 1: MPCU rear view



- (1) Ethernet address of MMC
- (2) X1
- (3) X2

- Now disconnect the following **front side** connectors:
 - Slot 2
X1: PCI-Link (It might be helpful also to disconnect the PCI-Link cable at the MC4C40 X4A and X2 as well as X4B at the GCTX. This gives you more room to pull out the MPCU).
 - Slot 4:
H1,H2: Physio data to PTC frontend
X1: CAN2
X2: CAN1
X3: Physio beeper to intercom
 - **I15**: Ethernet to host
 - **COM1**: Door contact
 - **COM2**: Physio graphic to host
- Unscrew the 4 screws "A" of the MPCU-Box shown in (→ Fig. 2 Page 6).

Fig. 2: MPCU front view



(1) Screws

- Pull out the MPCU-Box, be careful not to damage one of the cables.
- Disconnect the LED-board D20 K2203 at I14 .

1.2.3 Installing new MPCU1-Box



First write down the Ethernet address of MMC you find on a label at the rear side (→ Fig. 1 Page 5). You need this address for later configuration of NUMARIS!

- Put the LED-board D20 K2203 at I14 to the new MPCU-Box.
- Put the MPCU-Box back into the MMC frame.
- Tighten the 4 screws at the frame and reconnect all cables at front and rear side.
- Switch on the system again and check the 8 LED's at the D20 K2203.
 - » The mechanical work is done now.

1.2.4 Configuration of MPCU 133

(MPCU box part# 4753443, revision level 05)

1.2.4.1 Required tools

- Terminal or Laptop with terminal emulation software
 - 2 service cables PC-Host part # 5468947, Gender Changer part # 4757642 , Zero Modem Adapter part # 4680773. Connect both service cables to the Gender Changer. Connect the Zero Modem Adapter inbetween so you have the 9 pin plugs on each end of the assembled cable.
- Or: use standard RS232 Interface cable (9 to 9 pins).

1.2.4.2 Procedure

- Disconnect plug at COM 1 interface of MPCU and connect the laptop (Com1 port) or terminal via service cables.
- Setup of terminal/laptop: 9600 baud, 8 bit, no parity, 1 stop bit if you use Hyperterminal emulation, set properties:
Emulation: VT100
Connection: direct connection via COM1
- Reset the MPCU by pressing the Reset button on the IOP board D14.
- Interrupt the countdown of the MPCU by pressing any key on the keyboard.
- The following prompt will appear: (vxWorks Boot):
- Enter: c <RET>

After this input, you can step through the parameter list by pressing the <RET> key on your keyboard. Correct the settings as follows (delete any setting with a "."):

```

boot device                : dc
processor number           : 0
host name                  : HOST
file name                  :
inet on ethernet (e)       :
inet on backplane (b)      :
host inet (h)              :
gateway inet (g)           :
user (u)                   : vxworks
ftp password (pw) (blank = use rsh) : vxworks$1
flags (f)                  : 0x48
target name (tn)           : MPCU

```

startup script (s) :

other (o) :

- Enter: p <RET>
The new entries are displayed. Check your settings.
- After finishing the installation procedure of the netmask in Numaris, you can reboot the MPCU (enter : @ <RET> on your laptop) or reset the IOP again.
The MPCU now works with the new netmask.
- Shut down the system and power off/on to avoid system hang-up.

1.2.5 Software Configuration

The new Ethernet address of the MPCU1/2-Box must be known in NUMARIS.

Start the service software: Options -> Local Service

Home Menu -> Configuration (next)

Measurement -> Network (>)

MPCU MAC (input the Ethernet address without ":")

(Save and exit)

For further information refer to the on-line help and to technical documentation (→ Software Installation syngo MR A30 (NUMARIS/4VA30A) / MR-000.816.26).

1.2.6 Final Steps

Start NUMARIS and perform the complete MMC test in the Service Software, as well as a normal measurement using a phantom.

1.3 MPCU2/3/4-Box



The replacement requires approximately 30 minutes.

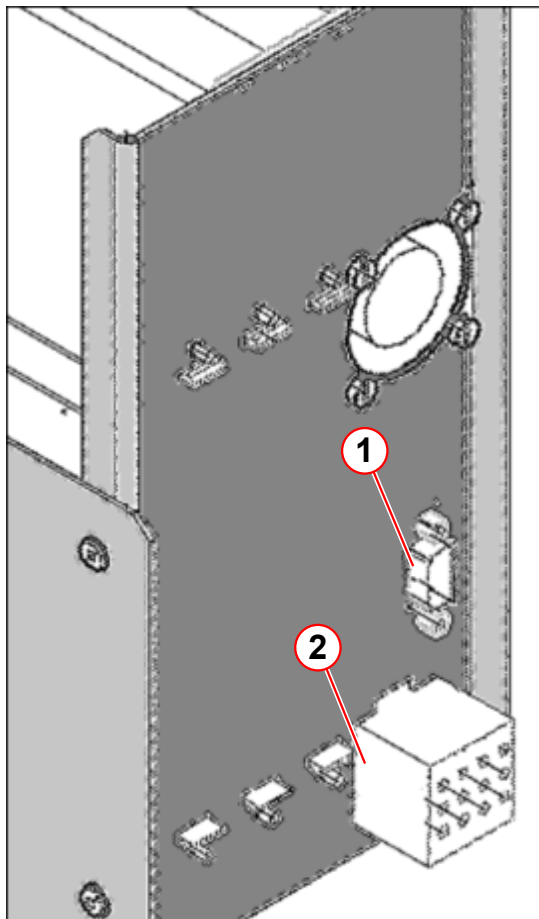
1.3.1 Preliminary Steps

- Shut down NUMARIS
- Switch off the system at the alarm box.

1.3.2 Removing old MPCU2/3/4-BOX

- First disconnect at **rear side** (→ Fig. 3 Page 9):
 - **X1**: MPCU Fan Control
 - **X2**: Power supply

Fig. 3: MPCU2/3/4 rear view

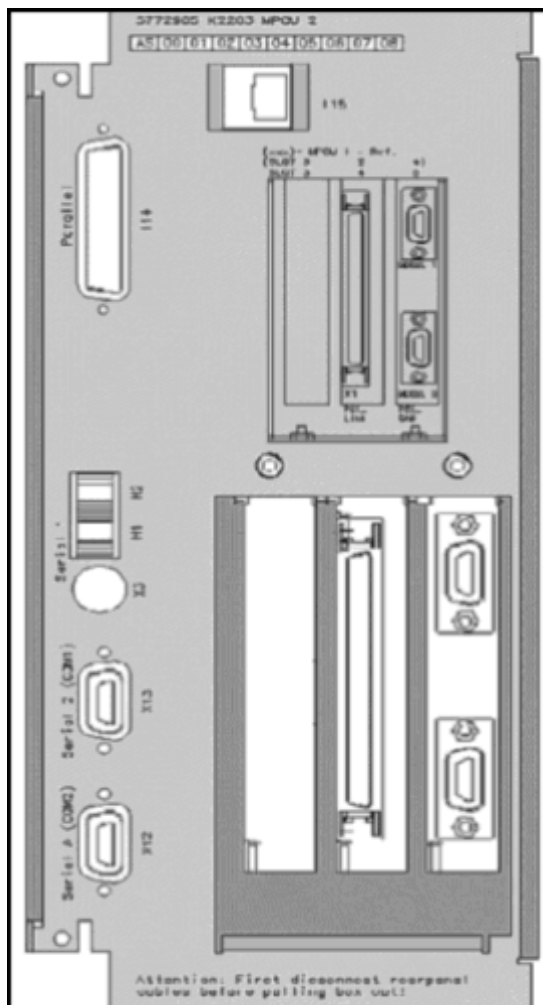


(1) X1

(2) X2

- Now disconnect the following **front side** connectors:
 - Slot 4:
 X1: PCI-Link
 - Slot5:
 PCI-CAN Interface (Module 0 and Module 1)
 - **MPCU2: H1,H2**: Physio data to PTC frontend (MPCU 3/4: COM2)
 - **MPCU2: X3**: Physio beeper to intercom (MPCU 3/4: COM2)
 - **I15**: Ethernet to host
 - **X13**: Door contact (MPCU3/4: COM1)
 - **X12**: RS 232C optional

Fig. 4: MPCU2/3/4 front view



- Pull out the MPCU2/3/4-Box, be careful not to damage one of the cables.
- Disconnect the LED-board D20 K2203 at **I14** .

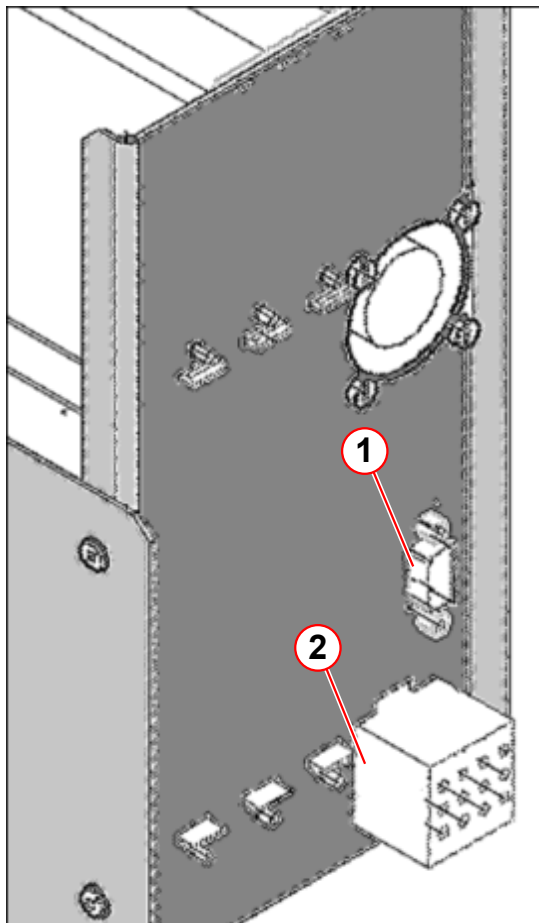
1.3.3 Installing new MPCU2/3/4-Box



First write down the Ethernet address of MPCU you find on a label at the front side (→ Fig. 6 Page 12). You need this address for later configuration of NUMARIS!

- Put the LED-board D20 K2203 at **I14** to the new MPCU2/3/4-Box.
- Put the MPCU2/3/4-Box back into the MMC frame.
- First connect the cables at **rear side** (Refer to (→ Fig. 5 Page 11)):
 - **X1**: MPCU Fan Control
 - **X2**: Power supply

Fig. 5: MPCU2/3/4 rear view



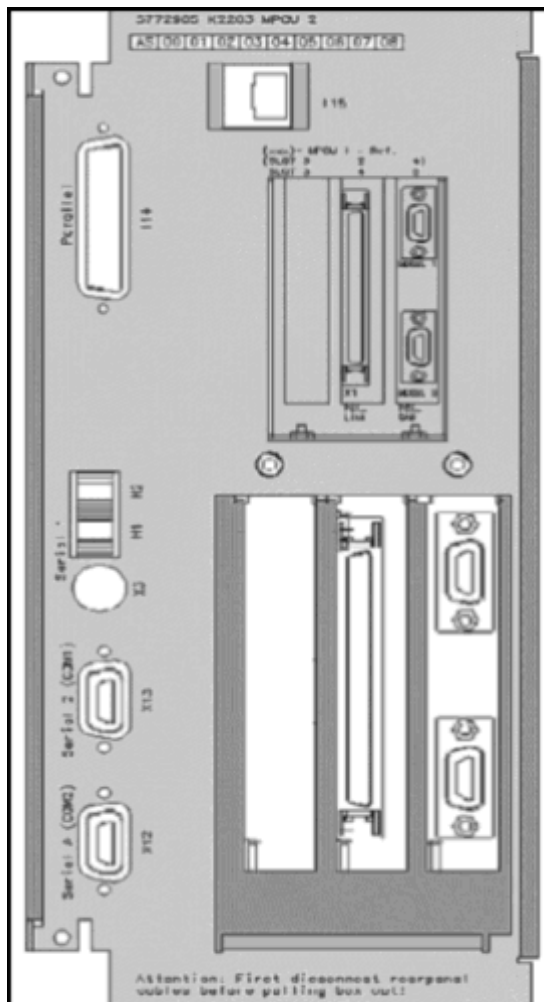
(1) X1

(2) X2

- Now connect the following **front side** connectors:
 - Slot 4:
X1: PCI-Link
 - Slot 5:
PCI-CAN Interface (Module 0 and Module 1)
 - **MPCU2: H1,H2**: Physio data to PTC frontend (MPCU3/4: COM2)

- **MPCU2: X3**: Physio beeper to intercom (MPCU3/4: COM2)
- **I15**: Ethernet to host
- **X13**: Door contact (MPCU3/4: COM1)
- **X12**: RS 232C optional

Fig. 6: MPCU2/3/4 front view



- Switch on the system again and check the 8 LED's at the D20 K2203.
 - » The mechanical work is done now.

1.3.4 Software Configuration



This configuration steps are not required any more if the installed software version is Numaris/4 VA25A or higher (e.g VA30A ...).

The new Ethernet address of the MPCU1/2/3-Box must be known in NUMARIS.

Start the service software: Options -> Local Service

Home Menu -> Configuration (next)

Measurement -> Network (>)

MPCU MAC (input the Ethernet address without ":")

(Save and exit)

For further information refer to the on-line help and to technical documentation "Software installation).

1.3.5 MPCU 4 software support



The MPCU 4 is only supported with software version Numaris/4 VB30A (syngoMR A30) with installed service pack 3.

Check if you have the required software version and service pack installed. Use the following procedure:

ACTION	MRC
■ In the main syngo MR UI, you can verify correct installation as follows: - Help -> Info - Check to make sure the "N4_VA30A_FEATURE_20070328_SP3_F11" entries are visible under the Siemens software packages. or check the log file C:\MedCom\MriProduct\inst\SWHistory.txt for the label entries "N4_VA30A_FEATURE_20070328_SP3_F11" If not, the software package is not installed correctly.	☐

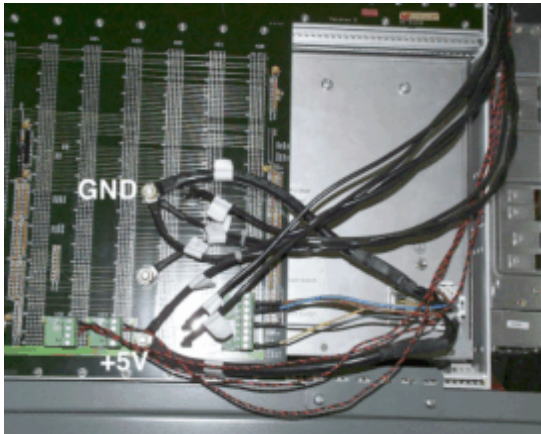
If not install the service pack 3 according the UI MR032/07/P Service Pack 3 for Numaris/4 VA30A.

1.3.6 Final Steps

Start NUMARIS and perform the complete MMC test in the Service Software, as well as a normal measurement using a phantom.

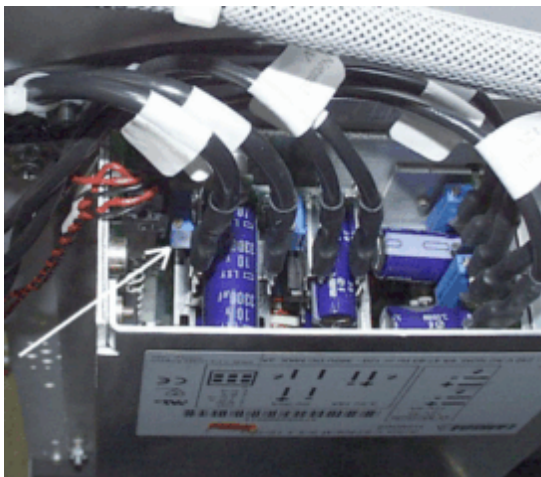
1.3.7 Adjustment of MMC power supply voltage

Fig. 7: Measurement of 5 VDC voltage at the MMC backplane



- In case of a replacement of the MMC power supply, adjust the +5 VDC supply voltage to 5.1 V with the pot on the power supply.

Fig. 8: +5 VDC adjustment potentiometer on the power supply



1.3.8 Replacement of MC4C40 and GCTX

To avoid timing errors, these boards have to be replaced together; (part no.: 73 84 444).

- » The common print number (MR-000.841.01) from Replacement of Parts is divided into separate Component print numbers.

Revision 02

- Hint: "Protective earth connection test" was added.

There are no Hazard IDs in this document.

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